10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

# Chapter 19 — Nitrogen and Sulphur

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

## Quick Checks (Solved)

### Quick Check 19.1

**1. Why is ammonia gas turned into a liquid after its production?**

The mixture coming out of the reaction chamber contains only about 35 % ammonia by volume, mixed with unreacted nitrogen and hydrogen. When this mixture is cooled by refrigeration coils, the ammonia alone changes into liquid ammonia at −33.4 °C, while nitrogen and hydrogen remain gases. In this way ammonia is easily separated from the mixture, the unused gases are recycled back into the chamber, and the liquid ammonia is easy to store, transport and use.

**2. Why is high pressure maintained in the production of ammonia?**

The reaction is reversible and the number of gas molecules decreases from four to two.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

According to Le Chatelier's principle, when the pressure is increased the equilibrium shifts towards the side having fewer molecules, that is towards ammonia. That is why a high pressure of about 200 atmospheres is maintained, which increases the yield of ammonia. High pressure also brings the gas molecules closer, so the rate of the reaction increases.

### Quick Check 19.2

**1. How is $SO_2$ prepared from $FeS_2$?**

Iron pyrite ($FeS_2$) is roasted, that is heated strongly, in excess of air. The sulphur present in the ore burns and sulphur dioxide gas is produced, while iron forms iron(III) oxide.

$$4FeS_{2(s)} + 11O_{2(g)} \rightarrow 2Fe_2O_{3(s)} + 8SO_{2(g)}$$

Sulphur dioxide can also be obtained by simply burning sulphur in air: $S + O_2 \rightarrow SO_2$.

**2. Why are $SO_2$ and oxygen purified before they react to produce $SO_3$?**

The gases obtained by roasting the ore carry dust and arsenic compounds with them. Arsenic compounds act as a catalyst poison and destroy the activity of the vanadium(V) oxide catalyst used in the next step. Therefore the gases are first passed through chambers where steam removes the dust and arsenic, and are then dried by spraying concentrated sulphuric acid in a drying tower. Clean and dry gases keep the catalyst active, so the conversion of $SO_2$ into $SO_3$ stays about 98 %.

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## Quick Check 19.3

### 1. Why lead oxide is called an amphoteric oxide?

An amphoteric oxide is one which reacts with both acids and bases to give salt and water. Lead(II) oxide behaves as a base in the presence of an acid and forms a salt, and it behaves as an acid in the presence of an alkali and forms a salt called a plumbate. Because of this double behaviour, lead oxide is called an amphoteric oxide. Zinc oxide and aluminium oxide behave in the same way.

### 2. Why non-metals form covalent oxides?

Non-metals have a high tendency to gain or share electrons and do not lose electrons easily. When a non-metal reacts with oxygen, which is also a non-metal, neither atom gives up its electrons completely. Instead, both atoms share electrons with each other, so a covalent bond is formed. That is why the oxides of non-metals such as $SO_2$, $CO_2$ and $NO_2$ are covalent compounds, and they are acidic in nature.

## Quick Check 19.4

### 1. Write the equation for the reaction between Ba and $H_2O$.

Barium is a reactive metal of group 2 and reacts vigorously with cold water, forming barium hydroxide and hydrogen gas.

$$Ba(s) + 2H_2O(l) \rightarrow Ba(OH)_{2(aq)} + H_{2(g)}$$

### 2. How do aluminum and zinc react with water?

Neither aluminium nor zinc reacts with cold water, because a thin protective layer of oxide is formed on their surface which prevents the reaction. Both of them, however, react with steam at high temperature and give their oxides and hydrogen gas. Aluminium reacts with steam only at about 700 °C, while zinc reacts moderately with steam.

$$2Al(s) + 3H_2O(g) \rightarrow Al_2O_{3(s)} + 3H_{2(g)}$$

$$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_{2(g)}$$

## Quick Check 19.5

### 1. Which will react faster with HCl, Zn or Fe?

Zinc will react faster with dilute hydrochloric acid. In the reactivity series zinc is placed above iron, so zinc is more reactive and loses electrons more easily. A more reactive metal reacts more vigorously with a dilute acid and gives hydrogen gas more quickly than a less reactive metal like iron.

### 2. Name one metal which will displace Zn from its salt.

Magnesium, or any metal placed above zinc in the reactivity series such as aluminium, calcium or sodium, will displace zinc from its salt solution. The more reactive metal pushes the less reactive metal out of its salt.

$$Mg(s) + ZnSO_{4(aq)} \rightarrow MgSO_{4(aq)} + Zn(s)$$

## Quick Check 19.6

**1. What are the harmful effects of oxides of nitrogen on human health?**

Oxides of nitrogen irritate the eyes, nose, throat and lungs, and cause coughing, breathing difficulty and asthma, especially in children and old people. They also form ozone and peroxyacetyl nitrate (PAN) in smog, which further damage the lungs. Nitric acid formed from them falls as acid rain, which makes the water and food acidic and causes respiratory and other health problems.

**2. How do oxides of nitrogen pollute environment?**

In the presence of sunlight, oxides of nitrogen react with unburned hydrocarbons and volatile organic compounds present in the air and form secondary pollutants such as ozone, aldehydes and peroxyacetyl nitrate (PAN), which are the main components of photochemical smog. They also react with water vapour and form vapours of nitric acid and nitrous acid, which fall as acid rain and damage crops, aquatic life, buildings and metals. In addition, they act as a catalyst in the oxidation of sulphur dioxide to sulphur trioxide, which then forms sulphuric acid in acid rain.

# Exercise (Solved)

## A. Multiple Choice Questions

**1. The reason for maintaining higher temperature for the production of ammonia is:**

(a) activation energy of the reaction is very high.
(b) activation energy of the reaction is very low.
(c) nitrogen and hydrogen are both gases.
(d) at low temperature nitrogen and hydrogen change into liquids.

**Answer: (a)** — The activation energy of the reaction between nitrogen and hydrogen is very high, so a high temperature is needed to give the molecules enough energy for a good rate of reaction.

**2. The contact process used for the industrial production of $H_2SO_4$ is sensitive to the impurities present in $SO_2$ and $O_2$ because the impurities:**

(a) affect the capability of the catalyst
(b) affect the purity of $H_2SO_4$
(c) do not let $SO_2$ react with oxygen
(d) decrease the rate of reaction appreciably

**Answer: (a)** — Arsenic compounds present in the gases poison the vanadium(V) oxide catalyst, so the gases must be purified before they enter the contact chamber.

**3. Which of the following oxides is neutral in character?**

(a) $Al_2O_3$
(b) $SO_2$
(c) $CO_2$
(d) NO

**Answer: (d)** — Nitric oxide (NO) does not react with water to form an acid or a base and has no action on litmus paper.

**4. Sodium is considered more reactive than magnesium because:**

(a) it is more electropositive than magnesium
(b) it reacts with water slowly.
(c) it is present in second group
(d) it is the less metallic

**Answer: (a)** — Sodium is more electropositive than magnesium, so it loses its electron more easily and is more reactive.

**5. Secondary pollutants present in the atmosphere are:**

(a) Oxides of nitrogen
(b) Oxides of sulphur
(c) Ozone and PAN
(d) Oxides of carbon

**Answer: (c)** — Ozone and peroxyacetyl nitrate are formed in the air from primary pollutants in the presence of sunlight, so they are secondary pollutants.

**6. $SO_3$ is absorbed in $H_2SO_4$ rather than $H_2O$ during the production of sulphuric acid because:**

(a) $SO_3$ does not react with $H_2O$
(b) reaction of $SO_3$ with $H_2O$ is highly exothermic producing mist of $H_2SO_4$ which is difficult to condense.
(c) it gives better yield of $H_2SO_4$
(d) reaction of $SO_3$ with $H_2O$ can cause explosion.

**Answer: (b)** — The reaction with water is highly exothermic and forms a fine mist of sulphuric acid which is very difficult to condense, so $SO_3$ is absorbed in 98 % sulphuric acid to form oleum.

**7. Oxides formed when oxygen reacts with metals:**

(a) Acidic
(b) Basic
(c) Both basic and amphoteric
(d) Neutral

**Answer: (c)** — Most metal oxides are basic, while oxides of some less electropositive metals such as zinc and aluminium are amphoteric.

**8. Major components of acid rain are:**

(a) $H_2SO_4$ and $HNO_3$
(b) $H_2SO_3$ and $HNO_2$
(c) $H_2SO_4$ and HCl
(d) Acetic acid and $HNO_3$

**Answer: (a)** — Sulphur dioxide and oxides of nitrogen form sulphuric acid and nitric acid in the atmosphere, and these acids fall with rain.

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## B. Short Answer Questions

### 19.1 How is nitrogen obtained from air?

Nitrogen is obtained by the fractional distillation of liquid air. First carbon dioxide is removed from air, and then the air is compressed to about 200 atmospheres and cooled. The compressed air is allowed to expand suddenly through a spiral jet, which cools it further; this process of compression and expansion is repeated until the air becomes liquid. The liquid air is then fractionally distilled. Nitrogen has a lower boiling point than oxygen, so it evaporates first at −196 °C, leaving oxygen behind. The nitrogen is then liquefied again and stored in special cylinders.

### 19.2 How is hydrogen produced from methane?

Hydrogen is produced by heating methane with a limited amount of oxygen, which is not enough to oxidise it completely to carbon dioxide and water. The products are mainly hydrogen and carbon monoxide. The carbon monoxide formed is then reacted with steam to give more hydrogen and carbon dioxide.

$$2CH_{4(g)} + O_{2(g)} \rightarrow 2CO(g) + 4H_{2(g)}$$
$$CO(g) + H_2O(g) \rightarrow CO_{2(g)} + H_{2(g)}$$

### 19.3 Which conditions are used to oxidize $SO_2$ to $SO_3$?

Sulphur dioxide is oxidised to sulphur trioxide by passing a mixture of clean and dry $SO_2$ and $O_2$ over a vanadium(V) oxide catalyst at a temperature of about 450 °C and a pressure of 2–3 atmospheres in a contact chamber. Under these conditions about 98 % of the sulphur dioxide is converted into sulphur trioxide.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

### 19.4 Why $CO_2$ is called an acidic oxide while CO is called a neutral oxide?

Carbon dioxide dissolves in water and forms carbonic acid, which turns blue litmus red and reacts with bases to form salts; therefore it is called an acidic oxide. Carbon monoxide, on the other hand, does not react with water to form an acid or a base, has no effect on litmus paper, and does not react with acids or bases; therefore it is called a neutral oxide.

| $CO_2$ — acidic oxide | CO — neutral oxide |
|---|---|
| Dissolves in water forming carbonic acid: $CO_2 + H_2O \rightarrow H_2CO_3$. | Does not react with water, so no acid is formed. |
| Turns blue litmus paper red. | Has no effect on blue or red litmus paper. |
| Reacts with bases to form salts, for example with lime water. | Does not react with either acids or bases. |

### 19.5 How do magnesium and calcium differ with each other towards their reactions with water?

Calcium is more reactive than magnesium. Calcium reacts vigorously with cold water and forms calcium hydroxide and hydrogen gas, whereas magnesium reacts with cold water only very slowly and gives magnesium hydroxide and hydrogen. To react properly, magnesium needs steam, with which it gives magnesium oxide and hydrogen.

| Calcium | Magnesium |
|---|---|
| Reacts vigorously with cold water. | Reacts only slowly with cold water. |
| Forms calcium hydroxide and hydrogen: $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$. | Reacts with steam to form magnesium oxide and hydrogen: $Mg + H_2O \rightarrow MgO + H_2$. |
| It is placed above magnesium in the reactivity series. | It is placed below calcium in the reactivity series. |

### 19.6 How are the reactivities of metals determined?

The reactivity of a metal is judged by the vigour with which it reacts with cold water, steam, dilute acids and oxygen. The more reactive a metal is, the faster and more violent is its reaction, more hydrogen is released and heat is produced. Reactivity is also judged by displacement reactions: a metal which displaces another metal from its salt solution is more reactive than that metal. By arranging metals in decreasing order of reactivity, the reactivity series of metals is obtained.

### 19.7 Which secondary pollutants are produced by oxides of nitrogen?

Oxides of nitrogen produce ozone ($O_3$), aldehydes and peroxyacetyl nitrate (PAN) by reacting with volatile organic compounds in the presence of sunlight; these are the main harmful components of photochemical smog. They also produce vapours of nitric acid and nitrous acid, which fall as acid rain, and they help in converting sulphur dioxide into sulphuric acid, which is another component of acid rain.

## C. Constructed Response Questions

### 19.1 Why ammonia is regarded as an important chemical?

Ammonia is one of the most important chemicals produced in the world. About 80 % of the ammonia produced is used to make nitrogenous fertilizers such as urea and ammonium salts, which are used to increase the production of food crops. It is also used in the manufacture of plastics and pharmaceuticals, and it is used as a refrigerant in refrigeration plants. In addition, about 13 % of the total nitrogen fixation in the environment takes place through the Haber process, so ammonia production plays an important role in the nitrogen cycle of nature.

### 19.2 How $SO_2$ present in the atmosphere is converted to $SO_3$?

Sulphur dioxide present in the atmosphere is slowly oxidised to sulphur trioxide by the oxygen of the air, and the oxides of nitrogen present in the air act as a catalyst for this reaction. The sulphur trioxide formed then reacts with water vapour and forms sulphuric acid, which falls to the earth as acid rain.

$$2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)} \quad \text{(catalysed by } NO_x\text{)}$$
$$SO_{3(g)} + H_2O(l) \rightarrow H_2SO_{4(aq)}$$

**19.3 The burning of fossil fuels in a car engine is responsible for the production of oxides of nitrogen. Explain.**

During the burning of petrol or diesel in a car engine, a very high temperature is produced in the cylinders. At this high temperature, the nitrogen and oxygen of the air react directly with each other and form nitric oxide. This nitric oxide leaves the engine with the exhaust gases and is slowly oxidised to nitrogen dioxide by the oxygen of the air. In this way the burning of fossil fuels in engines produces oxides of nitrogen in the atmosphere.

$$N_{2(g)} + O_{2(g)} \rightarrow 2NO(g)$$
$$2NO(g) + O_{2(g)} \rightarrow 2NO_{2(g)}$$

**19.4 The metals present at the top of the reactivity series are regarded as more reactive than those present at the bottom. Comment on this statement.**

The statement is correct. The metals placed at the top of the series, such as potassium, sodium and calcium, lose their electrons very easily. They react violently with cold water, catch fire in air and displace hydrogen violently from dilute acids. As we go down the series, metals such as iron, zinc and copper lose electrons with greater difficulty, so they react slowly with acids and do not react with cold water at all. Metals at the bottom, like silver and gold, are very unreactive. Hence reactivity decreases from top to bottom of the reactivity series, and the metals at the top are more powerful reducing agents.

**19.5 How fossil fuels produce $SO_2$?**

Fossil fuels such as coal and petroleum naturally contain sulphur compounds as impurities. When these fuels are burnt in factories, power stations and motor vehicles, the sulphur present in them combines with the oxygen of the air and forms sulphur dioxide, which escapes into the atmosphere with the flue gases. That is why fossil fuels are a major source of sulphur dioxide pollution and, in turn, of acid rain.

$$S(s) + O_{2(g)} \rightarrow SO_{2(g)}$$

## D. Descriptive Questions

**19.1 Explain the production of ammonia on industrial scale.**

Ammonia is produced on the industrial scale by the Haber process, which was discovered by a German chemist F. J. Haber. In this process nitrogen and hydrogen are mixed in the ratio of 1 : 3 by volume, heated to about 400–500 °C under a pressure of about 200 atmospheres and passed over an iron catalyst promoted with aluminium oxide.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \quad (\Delta H = -92 \text{ kJ mol}^{-1})$$

The nitrogen required is obtained by the fractional distillation of liquid air, and the hydrogen is obtained from methane. The equilibrium mixture leaving the reaction chamber contains about 35 % ammonia by volume. This mixture is cooled by refrigeration coils, when ammonia changes into liquid at −33.4 °C and is separated from the unused nitrogen and hydrogen, which are recycled back into the reaction chamber. In this way no raw material is wasted and a good yield of ammonia is obtained.

### 19.2 Describe the contact process for the production of sulphuric acid.

**(1) Preparation of sulphur dioxide:** sulphur dioxide is obtained either by burning elemental sulphur in air or by roasting iron pyrite ($FeS_2$) in excess of air.

$$S(s) + O_{2(g)} \rightarrow SO_{2(g)}$$
$$4FeS_{2(s)} + 11O_{2(g)} \rightarrow 2Fe_2O_{3(s)} + 8SO_{2(g)}$$

**(2) Purification of the gases:** the gases are passed through chambers where steam removes dust and arsenic compounds, and then through a drying tower where concentrated sulphuric acid removes moisture. This is necessary because arsenic poisons the catalyst.

**(3) Oxidation of sulphur dioxide:** the clean and dry mixture of $SO_2$ and $O_2$ is passed over a vanadium(V) oxide catalyst at about 450 °C and 2–3 atmospheres pressure in a contact chamber, when about 98 % of the sulphur dioxide is converted into sulphur trioxide.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

**(4) Absorption of sulphur trioxide:** sulphur trioxide is not absorbed directly in water, because the reaction is highly exothermic and produces a fine mist of sulphuric acid which is difficult to condense. It is therefore absorbed in 98 % sulphuric acid to form oleum ($H_2S_2O_7$), which is then diluted with the required amount of water to get sulphuric acid of the desired concentration.

$$SO_{3(g)} + H_2SO_{4(l)} \rightarrow H_2S_2O_{7(l)}$$
$$H_2S_2O_{7(l)} + H_2O(l) \rightarrow 2H_2SO_{4(l)}$$

Sulphuric acid is called the 'king of chemicals' because it is used, directly or indirectly, in almost every industry.

### 19.3 Explain the formation and properties of acidic and basic oxides.

**Basic oxides:** these are formed when metals are heated in air or oxygen. Metal oxides are usually ionic compounds in which electrons are transferred from the metal to oxygen. When dissolved in water they produce hydroxides, they turn red litmus blue, and they react with acids to form salt and water. Examples are sodium oxide, calcium oxide and copper(II) oxide.

$$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$$
$$CuO(s) + 2HCl(aq) \rightarrow CuCl_{2(aq)} + H_2O(l)$$

**Acidic oxides:** these are formed when non-metals such as sulphur, carbon and nitrogen combine with oxygen. Non-metal oxides are covalent compounds in which

electrons are shared between the atoms. These oxides react with water to form acids, they turn blue litmus red, and they react with bases to form salt and water. Examples are sulphur dioxide, carbon dioxide and nitrogen dioxide.

$$SO_{2(g)} + H_2O(l) \rightarrow H_2SO_{3(aq)}$$
$$CO_{2(g)} + 2NaOH(aq) \rightarrow Na_2CO_{3(aq)} + H_2O(l)$$

Some oxides, such as those of zinc and aluminium, are amphoteric and react with both acids and bases. Some oxides, such as CO and NO, are neutral and react with neither an acid nor a base.

### 19.4 Differentiate between first and second group metals based on their reactivities with oxygen and water.

| Group 1 metals (Na, K) | Group 2 metals (Mg, Ca) |
|---|---|
| They have one valence electron and are more electropositive. | They have two valence electrons and are less electropositive than group 1 metals. |
| They react very vigorously with cold water, giving hydroxide and hydrogen; potassium catches fire. | Calcium reacts vigorously but sodium is still more reactive; magnesium reacts only slowly with cold water. |
| They catch fire easily in air and burn with a characteristic flame; sodium gives sodium oxide and sodium peroxide. | They burn in oxygen with a characteristic flame, for example magnesium gives an intense white flame and forms MgO. |
| Their oxides dissolve in water to give strongly alkaline hydroxides. | Their oxides are basic; magnesium oxide is only slightly soluble, calcium hydroxide is partially soluble. |
| They are stored under kerosene oil to protect them from air and moisture. | They can be kept in air for some time, because a protective oxide layer forms on them. |

### 19.5 Describe the role of oxides of nitrogen in the formation of PAN and the acid rain.

**Formation of PAN:** during morning and evening traffic hours, vehicles and industries release a large amount of oxides of nitrogen ($NO_x$) and volatile organic compounds into the air. The ultraviolet radiation present in sunlight reacts with these gases through a series of complex reactions and produces secondary pollutants such as ozone, aldehydes and peroxyacetyl nitrate (PAN). PAN is a poisonous component of photochemical smog and it damages the eyes, lungs and plants.

**Formation of acid rain:** nitrogen dioxide reacts with the water vapour and other chemicals present in the air and forms vapours of nitric acid and nitrous acid. These acidic vapours mix with the water vapour, and when the water falls as rain its pH is less than 5.6; such rain is called acid rain. The oxides of nitrogen also act as a

catalyst and convert sulphur dioxide present in the air into sulphuric acid, which is another major component of acid rain. Acid rain corrodes buildings and metals, damages crops and aquatic life, and causes respiratory problems in human beings.

$$2NO_{2(g)} + H_2O(l) \rightarrow HNO_{3(aq)} + HNO_{2(aq)}$$

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